Peer or Steer: A Pilot Study Exploring Human-AI Collaboration in Creative Fields

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Recent years have brought immense advancements around development of Artificial Intelligence (AI) technology and its application. This has also led to a boost in interest on human-AI collaboration and co-creation. Specifically in the creative field, where it is usually not sufficient for an AI to solve given problems or produce deterministic output, this next-level interaction between humans and AI comes with huge potential but also major challenges, related to aspects such as role and power distribution, trust and reliance, or efficiency and effectiveness. In this paper, we present a pilot study on human-AI collaboration in three different creative fields (programming, marketing texting, and UI design), addressing User Experience, Technology Acceptance and, specifically, Perception of Collaboration. The study is based on a theoretical framework we derived from prior research through a focused, systematic literature review, and intended to raise research questions and identify related hypotheses informing future empirical work.

CCS Concepts: • **Human-centered computing** → **Interaction techniques**; **User studies**; *Interactive systems and tools*; • **Computing methodologies** → *Cooperation and coordination.*

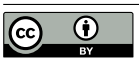
Additional Key Words and Phrases: human-ai collaboration, human-ai interaction, human-ai co-creation, collaborative ai

ACM Reference Format:
David Lang, Frederik Hirschmann, Karin Breckner, Thomas Neumayr, and Mirjam Augstein. 2025. Peer or Steer: A Pilot Study Exploring Human-AI Collaboration in Creative Fields. *Proc. ACM Hum.-Comput. Interact.* 9, 7, Article CSCW462 (November 2025), 34 pages. https://doi.org/10.1145/3757643

1 Introduction

Fueled by the recent rapid advancements in Artificial Intelligence (AI) technology, the idea of a "collaboration" between humans and AI tools has attracted drastically increasing interest over the past few years. This becomes obvious through the increase in related research recently published, which is easily exemplified by exploratory search for "human-ai collaboration" in

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ACM 2573-0142/2025/11-ARTCSCW462
https://doi.org/10.1145/3757643

titles of publications indexed on Google Scholar[1] as probably the broadest search system for academic literature, conducted in October 2024, which yielded 481 results in total, 435 of which had been published between 2021 and 2024 (and 475 since 2014). Despite this general boost in research interest, it must be noted that on one hand, the understanding of "collaboration" in this context, and even the understanding of "human-centeredness" is far from consistent across the scientific community. For instance, Capel and Brereton recently analyzed in existing literature what academic researchers understand by *human-centered AI* and found four major directions: explainable and interpretable AI, human-centered approach to designing and evaluating AI, ethical AI, and humans teaming with AI [8] (the latter being closest to our understanding of "collaboration"). Also literature explicitly claiming to research "human-AI collaboration" is highly diverse in the notion and understanding of collaboration (spanning from semi-supervised machine learning [50] to the communication with and characteristics of AI teammates [61, 62]). On the other hand, specifically the creative field, where humans and AI work together closely in order to co-create creative output (e.g., a user interface design, but also a marketing text or a program source code), poses a broad range of *potentials*: combining the complementary abilities of humans and AI (e.g., humans' creativity and emotional understanding and AI's capability to process large amounts of data and offer improvements based on learned patterns [57]) but also *challenges*: unintended effects due to biases, false sense of proficiency [7], unpredictability [33] or overconfidence [7].

This lack of clarity and consistency in the existing literature, as well as the specific challenges and potentials in the creative field, strongly motivated our research described in this paper. First, we analyze prior work through a focused, systematic literature review (see Sections 2 and 3), resulting in the identification of "core concepts" which we deem essential for a *human-AI collaboration* in our context. These *core concepts* then constitute the theoretical basis that in further consequence, informed design, methodology and analysis procedure of own study (see Section 4). We frame our study as *pilot study* – it has not been designed to lead to statistically reliable answers, but to manifold results produced by diverse participants solving diverse tasks in diverse creative fields (see Section 5), holding profound research questions and hypotheses as input to future research (see Section 6). Our main contributions thus are *1)* a small-scale yet systematic *review of the literature* and its analysis yielding *core concepts* of human-AI collaboration in related fields (such as role and power distribution, communication dynamics, or trust and reliance), *2)* a thorough *exploratory user study* addressing User Experience (UX), Technology Acceptance and Perception of Collaboration, and *3)* an in-depth *discussion* of its results including derivation of *hypotheses*.

2 Literature Review Methodology

In this section we describe our process of identification, evaluation and integration of relevant literature to provide a profound basis for our own research. We conducted the literature review systematically, thus we transparently report on search systems used, general search strategy, concrete search strings, and analysis methodology.

2.1 Databases

The selection of databases is crucial for conducting systematic reviews. For our review, we chose the ACM Digital Library and Scopus as search systems. Both, according to Gusenbauer and Haddaway, are well-suited as "principal search systems" for systematic reviews [17]. They meet essential criteria such as coverage, precision, and reproducibility. The *ACM Digital Library*[2] has a comprehensive collection in computing, including journals, conference proceedings, and technical magazines.

[1] https://scholar.google.com/, last access: 2024-10-28
[2] https://dl.acm.org, last access: 2024-10-24

This repository is useful for research that intersects with computer science, especially studies on algorithms, systems engineering, and especially human-computer interaction. *Scopus*[3] provides a broader perspective, covering a wide range of scientific journals across various disciplines, including technology, social sciences, and the arts. The multidisciplinary approach is essential for comprehensively capturing the diverse impacts of AI on various creative practices and for understanding the societal, ethical, and psychological dimensions of human-AI collaboration.

2.2 Search Strategy

A careful search strategy was used to navigate through the extensive literature, experimenting with different search strings to capture the essence and breadth of the topic at hand. We identified the following core concepts behind the topic of human-AI collaboration in creative fields as a basis for our research and the underlying literature review: *Collaboration*, *Creative*, *AI*, and *Interaction*. Table 1 explains the construction of our actual search strings on the basis of these core concepts: first, we identified synonyms, overarching terms and sub-concepts using dictionaries, glossaries, and preliminary (exploratory) literature review.

Core Concepts	Collaboration	Creative	AI	Interaction
Synonyms	Cooperation Teamwork Alliance Synergy Partnership	Inventive Innovative	Artificial Intelligence Machine Learning	Interface Interplay
Overarch. Terms	-	Creativity	Computer	Communication
Sub-Concepts	Co-Creation Multidisciplinary Teams Team Dynamics	Ideation Creativity Techniques Design Art Development Innovation Process	Intelligent Agents Deep Learning	Human-Computer Interaction HCI User Experience Usability

Table 1. Synonyms, overarching terms, and sub-concepts of core concepts that form the basis of our literature review.

2.3 Search Strings

The final search strings were constructed to align closely with the thematic cores of human-AI interaction, and collaboration in creative processes. To find scientific publications that not only mention these keywords somewhere in the text, a filter was applied to find results that contain the keywords in the abstract of the publication. Considering this topic is extremely volatile, we restricted the scope of the search to a time frame of ten years (thus, starting with publication dates in 2014). For Scopus, only specific subject areas such as *Computer Science*, *Social Sciences*, *Arts*, and *Psychology* were included to refine the search and ensure to focus on relevant domains. This contrasts with the search string for the ACM Digital Library, where no specific domains were excluded (due to its inherently narrower scope).

[3]https://www.scopus.com/, last access: 2024-10-24

ACM Digital Library query:

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{Abstract:(
   (Abstract: "collaboration" OR "cooperation" OR "teamwork" OR "alliance" OR "synergy" OR partnership" OR "co-creation")
   AND (Abstract: "AI" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "intelligent agents")
    AND (Abstract: "creative" OR "inventive" OR "innovative" OR "creativity")
    AND (Abstract: "interaction" OR "interface" OR "interplay" OR "human-computer interaction")
)} "filter": {ACM Content: DL}
```

Scopus query:

```
TITLE-ABS-KEY ( (
   ("collaboration" OR "cooperation" OR "teamwork" OR "alliance" OR "synergy" OR "partnership" OR "co-creation" )
    AND ("AI" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "intelligent agents")
    AND ( "creative" OR "inventive" OR "innovative" OR "creativity" )
    AND ( "interaction" OR "interface" OR "interplay" OR "human-computer interaction" ) ) )
AND PUBYEAR > 2013 AND PUBYEAR < 2025 AND ( LIMIT-TO ( EXACTKEYWORD , "Artificial Intelligence" ) )
AND ( LIMIT-TO ( SUBJAREA , "COMP" ) OR LIMIT-TO ( SUBJAREA , "SOCI" ) OR LIMIT-TO ( SUBJAREA , "ARTS" )
    OR LIMIT-TO ( SUBJAREA , "PSYC" ) )
```

2.4 Exclusion Criteria and Analysis

Our queries were conducted on 2024-02-12 (ACM DL) and 2024-02-27 (Scopus) and yielded 150 results in total (40 for ACM DL and 110 for Scopus). As shown in the PRISMA diagram [39] in Figure 1, we screened 146 documents after removing duplicates, and assessed 36 in detail for eligibility based on their full-texts, before finally including 28 publications for our review. Eight publications were removed based on the following Exclusion Criteria (EC): *Language: non-English language publication (EC1)*, and *Quality: non-scholarly publication (e.g., thesis proposals, workshop descriptions) (EC2)*. The identified 28 publications were then analyzed to extract core concepts relevant for human-AI collaboration in creative fields. The final body of literature and the core concepts extracted from it are presented in Section 3.

3 Related Literature

In this section, we discuss the related literature identified through the systematic literature review outlined in Section 2. From this corpus (see Table 2), we extracted thematic *core concepts* which we deemed highly relevant in the context of our own research (as listed in Table 3). In Sections 3.1 to 3.4, we discuss these core concepts in further detail. In Section 3.5, we then explain how the focused analysis of related literature has informed our empirical work.

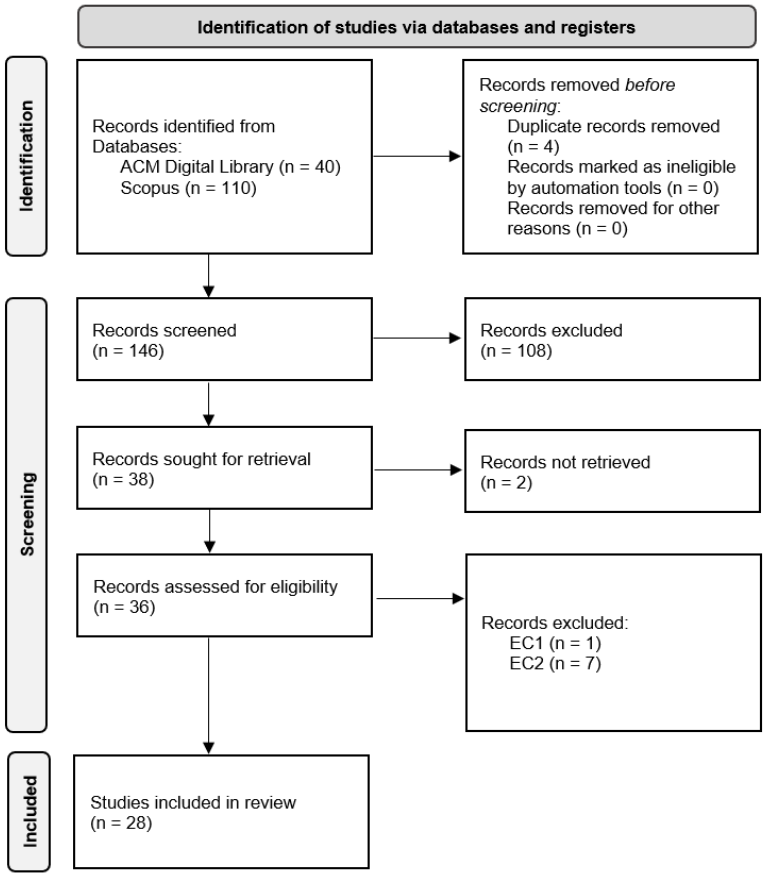


Fig. 1. Adapted PRISMA diagram for this literature analysis, based on [39]

Year	Author(s) [Reference]	Author Keywords	Publication Title
2024	Dodeja et al. [13]	Intelligent User Interfaces, Interactive Decision Support Systems, Design and Evaluation of Innovative Interactive Systems	"Towards the Design of User-Centric Strategy Recommendation Systems for Collaborative Human–AI Tasks"
2023	El-Zanfaly et al. [14]	Explainable AI, Graspable AI, Human-AI Co-Creation	"Sand-in-the-loop: Investigating Embodied Co-Creation for Shared Understandings of Generative AI"
2023	Ibarrola et al. [22]	Human-Computer Interaction, Computational Creativity, Image Generation	"A Collaborative, Interactive and Context-Aware Drawing Agent for Co-Creative Design"
2023	Lawton et al. [28]	Datasets, Neural Networks, Gaze Detection, Text Tagging	"When is a Tool a Tool? User Perceptions of System Agency in Human–AI Co-Creative Drawing"
2023	Lawton et al. [29]	Human-Computer Interaction, Cooperative AI, Design	"Drawing with Reframer: Emergence and Control in Co-Creative AI"
2023	Rezwana and Maher [43]	Human-AI Co-Creativity, Co-Creativity, Interaction Design, Framework	"Designing Creative AI Partners with COFI: a Framework for Modeling Interaction in Human-AI Co-Creative Systems"
2023	Shaikh and Cruz [48]	Artificial intelligence, Teams, Time, Interactions, Performance	"AI in Human Teams: Effects on Technology Use, Members' Interactions, and Creative Performance under Time Scarcity"
2023	Shi et al. [49]	Design, Artificial Intelligence, Literature Review	"Understanding Design Collaboration Between Designers and Artificial Intelligence: A Systematic Literature Review"
2023	Tchemeube et al. [52]	*no author keywords*	"Evaluating Human-AI Interaction via Usability, User Experience and Acceptance Measures for MMM-C: A Creative AI System for Music Composition"
2023	Vartiainen et al. [53]	Artificial Intelligence, Text-to-Image Generative Models, Generative AI, Midjourney, K-12, School Education, Prompt Crafting, Prompt Engineering	"Co-Creating Digital Art with Generative AI in K-9 Education: Socio-Material Insights"
2023	Zhang et al. [60]	Artificial Intelligence, Human-Centered Co-Creation	"Towards Human-Centred AI-Co-Creation: A Three-Level Framework for Effective Collaboration between Human and AI"
2022	Falk and Young [15]	Hackathons, Artificial Intelligence, Creativity Support Tool, Co-Creative Systems	"Supporting Fast Design: The Potential of Hackathons for Co-Creative Systems"
2022	Hwang [21]	Creativity, Human-AI Interaction, Creativity Support Tool	"Too Late to be Creative? AI-Empowered Tools in Creative Processes"
2022	Rezwana and Maher [42]	Co-Creativity, Interaction Design, AI to Human Communication, Human-AI Communication, Human-AI Creative Collaboration	"Understanding User Perceptions, Collaborative Experience and User Engagement in Different Human-AI Interaction Designs for Co-Creative Systems"
2021	Buschek et al. [7]	HCI, Artificial Intelligence, Co-Creation, Design	"Nine Potential Pitfalls when Designing Human-AI Co-Creative Systems"
2021	Davis et al. [11]	VR/AR/XR, Human-AI Collaboration, Creativity Support Tools	"Designing Co-Creative AI for Virtual Environments"
2021	Kim et al. [26]	Co-Creativity, Co-Creative System, Ideation, Collaboration	"Studying the Impact of AI-based Inspiration on Human Ideation in a Co-Creative Design System"
2021	Rezwana et al. [44]	Human AI Co-Creation, User Engagement, Virtual Embodied AI, Conversational AI	"Creative PenPal: A Virtual Embodied Conversational AI Agent to Improve User Engagement and Collaborative Experience in Human-AI Co-Creative Design Ideation"
2021	Wu et al. [57]	Creativity, Artificial Intelligence, Design Methods and Techniques, Design Process Management, HCI Theories and Methods, Education	"AI Creativity and the Human-AI Co-creation Model"
2021	Zhang et al. [59]	Storytelling, Oral Expression, Co-Creative System, Human-AI Collaboration, Creativity Support Tool, Children	"StoryDrawer: A Co-Creative Agent Supporting Children's Storytelling through Collaborative Drawing"
2020	Ashktorab et al. [3]	Games, AI, Agents, Collaboration, Social Perception	"Human-AI Collaboration in a Cooperative Game Setting: Measuring Social Perception and Outcomes"
2020	Louie et al. [33]	*no author keywords*	"Cococo: AI-Steering Tools for Music Novices Co-Creating with Generative Models"
2019	Andrews [2]	Evolutionary Art, Embodiment, DCog, Visualization	"An Embodied Approach to AI Art Collaboration"
2019	Guzdial et al. [18]	Artificial Intelligence, Human Computer Collaboration, Human AI Interaction	"Friend, Collaborator, Student, Manager: How Design of an AI-Driven Game Level Editor Affects Creators"
2018	McCaffrey and Spector [35]	Artificial Intelligence, Creativity and Ideation, Design Creativity, Human–Computer Interaction, Innovation, Man-Machine Systems	"An Approach to Human–Machine Collaboration in Innovation"
2018	Oh et al. [38]	Artificial Intelligence, Human Computer Collaboration, Human-AI Interaction	"I Lead, You Help but Only with Enough Details: Understanding User Experience of Co-Creation with Artificial Intelligence"
2017	Davis et al. [12]	Creativity, Interaction Dynamics, Collaboration, Qualitative Coding, Creative Agents, Drawing, Evaluation Methods	"Quantifying Collaboration with a Co-Creative Drawing Agent"
2017	Hoesl and Butz [19]	User Interface, Sense of Agency, Sense of Control, Sense of Authorship, Implicit Measures, Intentional Binding.	"Sense of Authorship and Agency in Computational Creativity Support"

Table 2. Our body of literature listed chronologically based on publication year.

Core Concepts	References
Roles, Control, and Power Dynamics	[7, 13, 13, 18, 19, 21, 29, 35, 38, 42, 52, 53, 60]
Communication Dynamics and Interaction Modality	[2, 14, 18, 25, 38, 43, 44, 49]
Trust, Reliance, and Social Perception	[3, 11, 30, 33, 38, 44, 59]
Creativity, Efficiency and Effectiveness	[12, 15, 21, 22, 26, 28, 29, 33, 42, 48, 49, 57]

Table 3. The core concepts we derived from the literature and related references (with multiple assignments possible, but without intention of completeness, since most of the publications at least peripherally touch most of these concepts).

3.1 Roles, Control, and Power Dynamics

In a *creative collaboration* between two individuals, the resulting creative outcome cannot be attributed to one party alone, according to Rezwana and Maher. It is always a contribution from both parties, surpassing the original intentions of each individual, which is fully applicable to human-computer collaboration [42] but also generally applies for other forms of "collaboration". Dodeja et al. particularly emphasize the importance of *shared goals* when collaborating with an AI. It is essential that users can express their intentions, and that the AI is able to comprehend them.

Further, and specifically related to *roles* in a human-AI collaboration, Hwang in [21] mentions both *role distribution* and *individual personality* as significant: *Should AI work for or with the human?* Hwang proposes that users "should hold the autonomy in deciding whether they would prefer AI to work *with* or work *for* them" and further suggests that such *role assignments* should be part of the "initial parameters to kick-start co-working processes with AI" [21]. The research of Guzdial et al. further analyzes *role dynamic* by asking study participants to mentally put an AI into various roles such as friend, collaborator, student, or manager. They found these mental models to influence users' interactions in different ways, e.g., a participant verbalized their expectations of AI as collaboration partner complaining that the AI "didn't seem to be trying to create any consistency with my initial choices as you might expect from a human collaborator" [18]. This suggests that also correctly framing the capabilities of an AI is important because human expectations align with their mental model of what the AI should be able to do. This again is indirectly supported by the findings of Dodeja et al. who found a significant correlation between the (perceived) usability of an AI system and the impression of intelligence it conveys [13]. Also related to *role dynamic*, Lawton et al. found users to toggle between tool-like use and agentive interaction depending on what they wanted to achieve in a specific moment [29]. Yet, in the research of Oh et al., users didn't like it when the AI fluctuated between human- and AI-like behavior, they preferred the AI to act predictably human-like, although in the creative process they also benefited from unexpected results of the AI [38].

A major challenge when incorporating AI into everyday creative processes, strongly related to *power distribution*, is to ensure that AI systems integrate seamlessly into existing workflows without disrupting the creative flow. Moreover, using AI extensively in creative efforts raises ethical questions, including concerns about intellectual property rights and the risk of losing the personal touch in artistic works. To address these difficulties, Zhang et al. in [60] highlight the need for a collaborative approach, where human creatives and AI systems work together harmoniously. By leveraging the strengths of both parties, it becomes possible to unlock new forms of innovation and creativity while overcoming challenges. Hwang in this regard, argues that a large portion of AI tools in the creative field "targets *non-designers*, reducing the risk of depriving a user's creative role", but immediately adds that because creativity is omnipresent in today's work content, employees

who are not professional designers or creators may also be "heavily involved in producing creative work". Thus, according to her, the "push-back" from human users will not fully be resolved [21].

In this context, related literature also discusses *authorship*, indicating that users prefer to remain feeling as "authors" of the creative output they produced together with an AI. Not providing this sense of authorship to the users inherently holds a risk for negatively affecting UX, similarly this applies for the sense of *control* and *self-agency* [19]. Relatedly, the study of Vartiainen et al. shows that the sense of authorship can vary drastically among users: part of their participants in an identical study setting considered themselves to be (co-)*creators* of the output they produced using a generative AI, while others did not [53]. The research of Tchemeube et al., who investigated an AI tool for co-creative music composition regarding usability, UX and acceptance by composers, further found that limited control over AI-generated outputs negatively impacts users' sense of creative ownership [52].

Additionally, and related to the specific dynamic of roles and power distribution in a human-AI collaborative setting, *overconfidence* in the capabilities of the AI is a potential source of problems. AI can suggest things that do not fit the current context or are simply not true. For users, it is easy to believe something if it seems to be high quality output. According to Buschek et al., AI is well capable of making something false seem real because it presents it in high quality, e.g., well-written sentences [7]. This effect is also included in Buschek et al.'s nine pitfalls in design of human-AI co-creative systems, e.g., phrased as "false sense of proficiency" or "invisible AI boundaries" [7]. As explained by Zhang et al., related effects can negatively impact humans' perception of their own significance in the collaborative process: "[A]s AI takes increasing agency in creative works, it may cause risks of undermining humans' self-efficacy in collaboration." [60]. According to the findings of McCaffrey and Spector, it is thus important that, in order to allow for innovative collaboration, human and AI can *complement each other* [35]. McCaffrey and Spector further point out that such a collaboration can help overcome cognitive biases or functional fixedness.

3.2 Communication Dynamics and Interaction Modality

Rezwana and Maher highlight that many human-AI co-creative systems lack a mechanism for two-way *communication*. Their analysis of 92 co-creative systems showed that the majority depended exclusively on human-to-AI communication, without allowing AI to communicate back to users. Rezwana and Maher conclude that "AI can only communicate by contributing to the product and does not directly communicate to humans" [43]. They identify this "gap in interaction design" as a major area for future research in the field of co-creativity. Along a similar line, Andrews emphasizes two-way communication as essential for collaboration [2]. Oh et al. further point to the importance of *context-awareness* for an AI's communication and argue that users appreciate feedback and specific commentary, particularly if context-sensitive to their work or intentions [38]. Similarly, the research of Guzdial et al. shows that users in the field of design valued AI that learns from their inputs and refines its suggestions accordingly, enhancing its utility in the creative process [18]. Relatedly, Rezwana et al. found that it helps to let the users individualize the output they want to get from the AI [44].

Interaction modality is crucial for any form of *communication*. In our context, it refers to the types of input and output used, exploring the channels for exchanging information between humans and AI. Identifying these channels is the first step towards exploring the depth and potential of collaborative efforts [25]. Also Shi et al. stress the significance of *interaction modality* for adequate support of collaboration. In their literature review, they found that specifically in designer-AI collaboration, communication barriers caused by limited input modalities might prevent designers from giving their ideas to the AI and thus hinder effective collaboration [49]. Specifically, they present this aspect as an implication for future research and practice based on findings of

earlier studies, not as a concrete limitation of today's AI tools. However, they point to concrete communication issues caused by AI tools generating output without providing explanation. It can be expected that the current trend towards explainable AI contributes to making state-of-the-art AI tools better address these barriers. Rezwana et al. reinforce the relevance of this aspect by stating that the design of the interaction is crucial for fostering a good collaboration in human-AI teams.

While the most prominently discussed interaction modalities in the creative field include visual, linguistic, gestural, and touch [49], there are also examples for unconventional ones such as the one described by El-Zanfaly et al., where a sand playground was used as interactive setting and proved effective in fostering engagement and collaboration [14].

3.3 Trust, Reliance, and Social Perception

Trust is a crucial aspect in human-AI collaboration. In their (more general) research on trust in automation, Lee and See point out that "[a]utomation is often problematic because people fail to rely upon it appropriately" and because "people respond to technology socially, trust influences reliance on automation" [30]. This also hints that trust might heavily influence future behavior and usage patterns of individual users. In the context of trust in collaboration and shared creativity in human-AI settings, *unpredictability* is oftentimes considered problematic. Accordingly, the findings of Louie et al. suggest that *trust* and *reliance* (but general sense of collaboration) are positively affected by enhanced control possibilities for users [33]. Similarly, Oh et al. argue that users appreciate *transparency* related to the mechanics behind the AI [38]. They compare it to behavior in human teams where team members usually want to understand why the other person(s) handled a situation or input in a specific way, and explain to each other their intentions. On the other hand, Davis et al. think of *unpredictability as a chance* for the users to think out-of-the-box. They argue that systems which "wish to encourage collaborative machine creativity should embrace the unpredictable nature of these systems as a source of inspiration and unexpected ideas" [11]. Similarly, Oh et al. report that in their study, although humans preferred *predictable*, human-like AI behavior, they also benefited from unexpected results in the creative process [38]. While the different lines of argumentation prevalent in the recent literature on human-AI collaboration in creative fields that is part of our review, seem to be contradictory, they seem to greatly match with earlier discourse on the role of (un)predictability for creativity. For instance, Boden explicitly addresses unpredictability in the context of creativity, and describes it as "the essence of creativity" [5]. Although she emphasizes that "creativity is not enough", she concludes that "[c]onstraints and unpredictability, familiarity and surprise, are somehow combined in original thinking" [5].

Another relevant dimension strongly related to *trust* is *social perception*. In a study described by Ashktorab et al. investigating AI in gameplay settings, participants consistently rated AI partners lower in aspects related to social perception than their human counterparts, despite identical performance results [3]. This discrepancy reveals a bias that potentially impacts the (perceived) quality of human-AI collaboration and aligns with the universally desirable goal to create AI systems that convey a good UX. Further, this effect again relates to Buschek et al.'s nine potential pitfalls when designing human-AI co-creative systems, which, for instance, suggest that users' *perception* of an AI's proficiency affects both *trust* and *reliability* [7].

One approach to increasing humans' *trust* in AI is *embodiment* which, according to Rezwana et al. can help to engage users more and facilitate the impression of not just using a tool. In their study, Rezwana et al. introduced a character (PenPal), which looks just like a pen, as the creative AI partner in a sketching environment. The affective expressions of the character helped users to foster a relation to it [44]. Embodiment can potentially also help to better fit the mental model of people of different age. For instance, the research of Zhang et al. suggests a difference in the *perception* of AI for differently aged kids: in a learning environment, the younger children felt the

AI to be more of a friend or compared it to their favorite teacher, while older children viewed the AI more as a tool [59].

An aspect oftentimes associated negatively with *trust* in AI systems is *bias* (or unfairness), e.g, caused by biases already present in training data. Buschek et al. exemplify undesirable outcomes such as gender stereotypes [7]. In the context of creative work, these stereotypes might influence development of content for a specific gender, or cause unintended effects when creating characters. It is thus crucial for users to be aware of potential biases and to address them. This can be challenging due to factors such as overconfidence, or lack of transparency regarding these biases.

3.4 Creativity, Efficiency, and Effectiveness

Related to *creativity* in our research context, Davis et al. distinguish between two different approaches: *computational creativity* and *creativity support* tools. Computational creativity implies that AI systems can autonomously produce innovative outcomes, whereas creativity support tools are designed to stimulate and enhance human creativity [12]. In the context of human-AI collaboration, *creativity* is oftentimes described as *co-creativity*. Falk and Young define co-creativity as humans and AI agents collaboratively creating shared artifacts [15], similarly, Rezwana and Maher describe it as a creative process involving humans and AI as partners to produce creative output [42]. Many authors emphasize the *diverse* and *complementary* potentials that humans and AI can contribute to a creative collaboration. For instance, in Wu et al.'s human-AI co-creation model, humans contribute creativity, emotional understanding, and subjective judgment, while AI excels at processing large volumes of data and offering improvements based on learned patterns [57]. They envision the interaction as highly iterative, with AI providing suggestions and feedback to refine human ideas continuously. Over time, AI learns from human preferences, better aligning with the creator's needs and tastes, thus enhancing *creativity* and *efficiency*. This is in line with Lawton et al.'s research, identifying *collaboration* as an important driver for *creativity* – from software development to musical improvisation [28].

According to Shi et al., a factor that promotes collaboration is its *scope*. They argue that a broad scope can facilitate more effective collaboration [49]. This implies that if an AI is capable of assisting throughout a process, rather than merely performing a single task, it can enhance *creativity* and overall *efficiency*. A potential negative effect of using AI in creative teamwork on the other hand is described by Shaikh and Cruz. Their research suggests that especially when time constraints are present, involvement of AI can lead to a reduction in interaction between the human team members, which subsequently affects the team's overall creativity [48]. Further, Lawton et al. suggest that the *effectivenes* of collaboration with an AI-tool can also vary considerably due to the nature of the task at hand [29]. Using the wrong tool for the wrong task may lead to a non-effective collaboration and also may inhibit further acceptance of a tool.

Finally, the general range of *creative disciplines* where AI could play a role is huge. One prominent example that is specifically interesting in the context of human-AI co-creation, is the broader field of *design*, which is addressed in many of the publications in our body of literature (e.g., [21, 22, 26, 28, 29, 44, 49, 59]). This field deserves such high attention because designers typically do not know exactly what to look for right at the start of the creative process [22]. Thus, according to Ibarrola et al., "it is not sufficient to focus on the problem of generating useful, novel or diverse proposals" (although, e.g., the research of Kim et al. confirmed the importance of AI providing high-quality and conceptually relevant inspirations [26]), but we need to "see the task as a collaborative user/system process [characterized] by both sides developing, refining, and aligning their intentions and adapting to each others' changing goals" [22]. Given the discourse on lack of two-way communication in co-creative systems and the related research gap identified by Rezwana and Maher (see above), this

hints a potential discrepancy between demand and current capabilities of AI systems, and makes design a particularly interesting research domain in our context.

3.5 Conclusions

The findings from the literature included in our focused systematic review, greatly informed our own study (see Section 4) in several ways: on one hand, related to *task selection*, *tool selection*, but also *selection of participants* (which affects general *study design*), and related to *research methodology* and *instruments* used. Regarding the former, we initially compiled a large set of tools, carefully analyzing and testing them for our purposes before making a selection. We considered aspects such as task-tool fit, scope of the tools, and relatedly also the diversity of the tools finally included. Similarly, we designed the three tasks to align with different requirements for co-creative AI tools in different fields (specifically, including one from the field of *design*). Further, in order to account for potential differences between users with different levels of experiences, we strove for involving participants with diverse backgrounds. Additionally, to avoid unintended/undesired effects in human-to-human team dynamics, we reduced the scope of our study to human-AI constellations involving one human only. Regarding methodology, we have added to the more general and well-established instruments for the analysis of UX and Technology Acceptance, items that allow for investigation of our identified *core concepts*, such as *role dynamics*, *trust and reliance*, *interaction dynamic* or *goal alignment*.

4 User Study

In this section we describe the research methodology followed by our pilot study (see Section 5 for the results) to explore the key factors that contribute to collaboration in human-AI interactions within creative fields. It introduces the selected tools, our study procedure, the data collected, tasks to be performed during the study, and participants.

4.1 Selected Tools

The AI tools employed for our study were selected carefully and systematically based on the following criteria: *relevance* for the concrete setting (i.e., for creative tasks), *interaction modality* (e.g., natural text, voice commands, graphical input), *availability* (i.e., free of cost or commercially available, price range), *complexity* (i.e., intuitiveness and user-friendliness) and *functional range* (i.e., whether a tool has a broad, generic or a specialized range of functions). For the study we selected tools that 1) are well suited for creative tasks, 2) cover different interaction modalities (natural-text-based, code-text-based and GUI-based interaction), 3) are free of charge or low-cost, 4) intuitive to use (so that no lengthy introduction was necessary for first-time users), and 5) cover both, specialized and generic set of functions. Besides the listed criteria, we referred to Hwang's clustering system using four different categories of co-creative tools: editors (i.e., facilitating easy content editing), transformers (i.e., facilitating conversion of content from one form to another), generators (which produce ready-to-use creative outputs based on user inputs), and blenders (which combine multiple creative elements to generate new outputs) [21]. In total, we considered a high number of 51 tools (listed in Tables 8 and 9 in the Appendix) and finally selected three of them best matching the requirements of the three different tasks described below in Section 4.4. We briefly describe them in the following.

GitHub Copilot. Copilot[4] was chosen for a programming task. It is an AI-powered *coding assistant* that has been developed by GitHub[5] and OpenAI[6]. It runs in the developer's IDE and *suggests code* directly or *auto-completes* code. It supports a multitude of different coding languages and has been designed with the intention of providing assistance to programmers, which allows them to focus more on the resolution of issues [16]. This approach aligns with the coders' workflow as there is no requirement for them to adapt to an unfamiliar environment. Furthermore, it is an interesting tool because of the nature of its interaction modality, which places emphasis on comprehension of the code. According to Hwang's categorization [21], GitHub Copilot is an example for a *generator*.

ChatGPT. ChatGPT[7] (using GPT-4) is a state-of-the-art large language model developed by OpenAI. It is known for its ability to *generate human-like text* based on the input it receives. GPT-4[8] significantly improves upon its previous versions in both scale and capability, offering a more nuanced understanding and generation of text. It can perform a *wide range of tasks*, including but not limited to writing essays, summarizing content, translating languages, and generating code. Its enhanced performance is attributed to its training on a diverse and expansive dataset, allowing it to provide more accurate and contextually relevant responses across various domains. ChatGPT can currently be considered the most used AI-tool on the market, according to [51]. ChatGPT offers an *easy-to-use text-based interface*. Its output can be used in a variety of creative fields. According to Hwang's categorization [21], ChatGPT is an example for a *generator*. In our study it was used for a creative task in the *marketing* domain.

Uizard. Uizard[9] developed by Uizard Technologies[10], a Copenhagen-based company specializing in AI-powered design tools, is a tool helping *create screen designs* and is therefore highly relevant for UI/UX designers. While it has a very *narrow field* regarding the creative domain it is made for, its variety of UI/UX tasks is rather wide. E.g., it can be used to *generate a complete project* including several screens, or to *make slight variations of designs*. According to Hwang's categorization [21], Uizard can be considered a *transformer*.

4.2 Apparatus and Procedure

Given that all tasks (see Section 4.4) could be completed solely on a computer screen and no additional hardware (or software that needed to be installed) was necessary, we decided to conduct the study remotely. This allowed the participants to stay in their familiar environment (e.g., their workplace or home-office). The study was conducted by one main researcher who was connected to the individual participants via a Zoom[11] meeting during the whole duration of the study session, in order to collect additional data such as facial expressions or verbal comments, and to potentially answer arising questions. Although Zoom generally allows for multiple screen sharing[12], this was not possible in the concrete setup. Users were thus not allowed to use a second screen, except for displaying the task handout (in case they had not printed it). While this was not a problem with two of the tools (GitHub Copilot and Uizard), where the AI is built into the tool, for ChatGPT, users

[4]https://github.com/features/copilot, last access: 2024-10-28
[5]https://github.com/, last access: 2024-10-28
[6]https://openai.com/, last access: 2024-10-28
[7]https://openai.com/chatgpt,last access: 2024-10-28
[8]At the time the study was conducted, GPT-4 was the most recent model.
[9]https://uizard.io/autodesigner, last access: 2024-10-28
[10]https://uizard.io/, last access: 2024-10-28
[11]https://zoom.us, last access: 2024-10-25
[12]https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0064237, last access: 2024-10-11

were asked to use a split screen with ChatGPT in the browser on one side and the text editing tool on the other. To prove this setting to be effective, several thorough pre-tests have been conducted.

The actual study sessions began with a brief orientation to introduce participants to the goals of the study, the tasks at hand, and the AI tool involved. After the introduction, users were asked to complete and sign a consent form, confirming that they were willing to take part in this research and agreed to the use of the data collected (compliant with GDPR). Given the varying levels of familiarity with AI technologies, a brief explanation of the tool was provided if necessary. Users were further handed out a one-page task description and explanation. During the actual task execution which followed directly after, participants were free to ask questions, further enriching the data with their queries and concerns. Upon completion of the tasks, participants were asked to complete a comprehensive questionnaire (see Section 4.3), and subsequently interviewed individually in an open manner. These brief interviews were conducted in order to be able to ask the participants individually-tailored questions based on aspects that had been observed before. For instance, they comprised questions related to rationales behind a participant's decisions and behavior, or their feelings in specific situations. The interviews were not intended as a general additional data source but to confirm observations and to clarify the observer's (potentially) remaining open questions. They thus did not follow a structured guide, but differed for every participant.

4.3 Data Collection and Instruments

Data collection primarily focused on the participants' answers to the comprehensive *questionnaire*, filled in individually after completing the task. The questionnaire was hosted via Google Forms[13] and could be accessed via a link or QR-code sent to participants and placed on the handout. Additionally, the whole interaction process was *observed* by one main researcher, and participants were asked to *think aloud* during the process if they felt comfortable with doing so. The observer further took *notes* on participants' comments and general behaviour during the interaction process. The whole process including the brief interview in the end, was *screen-, video- and audio-recorded*, to capture the interactive dynamics between the participants and the AI tools, and to facilitate later investigation as necessary. The recordings, observational data, and notes allowed for additional insights related to the interaction process complementing the rich data gathered via the questionnaire (e.g., related to what interaction modalities participants used, how often they consulted the AI, and whether they used or disregarded the AI's output).

In the following, we further describe our comprehensive *questionnaire*, comprising 59 items in total. During a pre-test, we recorded an average completion time of 17 minutes for the questionnaire, which is relatively lengthy. However, since the questionnaire was essential, and the major instrument for our study, we decided to accept a slightly longer completion time in order to ensure comprehensive coverage of relevant content. The questionnaire was structured in four distinct parts: *general information*, *UX*, *Technology Acceptance*, and *Perception of Collaboration*, as detailed below.

General Information. The first part of the questionnaire included ten items related to participants' demographic and professional background, comprising age, gender, current profession, years the person had worked in this profession, prior experience using AI systems in general, concrete AI systems the participant had been using before, and prior experience with the AI tool that the participant used in the study (i.e., either GitHub Copilot, ChatGPT or Uizard). In case the participant had prior experience with this particular tool, we further asked for tasks the tool had been used for and frequency of use. For the individual questions, we used different scales as applicable, or free text to be entered. For instance, we used 5-point Likert scales for questions related to prior

[13]https://docs.google.com/forms, last access: 2024-10-21

experience, or a single choice list with the options established in our country for gender. Free text questions were, e.g., related to profession or AI tools used before.

User Experience. This part comprised several modules of the standardized User Experience Questionnaire+ (UEQ+)[14] [46, 47], a modular extension to the UEQ[15][27]. The UEQ (as well as UEQ+) comprises various subscales: six in the non-modular original UEQ [27], 27 in total for the UEQ+ [46, 47], which are however intended to be individually selected and tailored to the concrete scenario. Each subscale comprises several items in semantic differential format (a 7-point Likert scale between opposing pairs of product properties such as complicated vs. easy). For our questionnaire, we selected five of the subscales, containing four items each, resulting in 20 items in total: *Efficiency (EF)* (e.g., slow vs. fast or inefficient vs. efficient), *Perspicuity (PE)* (e.g., difficult to learn vs. easy to learn or confusing vs. clear), *Dependability (DE)* (e.g., unpredictable vs. predictable or obstructive vs. supportive), *Trust (TR)* (e.g., unreliable vs. reliable or untrustworthy vs. trustworthy), and *Intuitive Use (IU)* (e.g., difficult vs. easy or inconclusive vs. conclusive).

Technology Acceptance. This part integrated a relevant subset of the Technology Acceptance Model 3 (TAM3) [54]. The original TAM in its first version [10] comprised twelve items (using 7-point Likert scales) related to perceived usefulness and and perceived ease of use. The most recent version TAM3 [54] has a considerably larger scope, including 51 items in 16 subscales (in addition to the two original ones, e.g., computer self-efficacy, computer anxiety, job relevance or result demonstrability). Part of the subscales of the TAM3 are not relevant in the context of our study, others strongly overlap with other parts of our questionnaire. Based on these considerations, we selected a subset of TAM3, including five subscales with 18 items in total: *Perceived Usefulness (PU)* (four items, e.g., "using the system improves my performance in my job"), *Perceived Ease of Use (PEOU)* (four items, e.g., "interacting with the system does not require a lot of my mental health"), *Behavioral Intention (BI)* (three items, e.g., "given that I had access to the system, I predict that I would use it"), *Computer Playfulness (CPLAY)* (four items, e.g., "spontaneous" or "creative" as response to "characterize yourself when you use computers"), and *Perceived Enjoyment (ENJ)* (three items, e.g., "the actual process of using the system is pleasant").

Perception of Collaboration. Although the combination of UEQ+ and TAM3 constructs already includes a considerable number of standardized questions, both instruments were not designed with the specific objective of measuring perceptions of collaboration. Consequently, we developed a series of additional items and integrated them into the questionnaire. These questions were derived from the findings presented in Section 3, and revolve around the concepts and perception of collaboration (e.g., related to trust and reliance, alignment of goals and power distribution). In total, this part of the questionnaire included eleven items in eight dimensions: *Trust and Reliance (REL)* (one item: "I felt confident about the AI's abilities"), *Transparency (TRA)* (one item: "I understood how the AI made its decisions"), *Effective Communication (EC)* (two items: "the communication between me and the AI was effective" and "the AI's communication was relevant"), *Aligned Goals (AG)* (one item: "I felt my goals and the goals of the AI to be aligned"), *Role Distribution (RDIS)* (two items: "the distribution of roles between me and the AI was effective for the task" and "looking at the result of the task, who contributed more?"*), *Power Distribution (PDIS)* (two items: "I felt in control of the decisions made during the task" and "the decision-making power seemed balanced between me and the AI"), *Interaction Dynamic (IDYN)* (one item: "describe how you and the AI worked together, was it more collaborative or directive?"*), and *Self-Efficacy (SEFF)* (one item: "I felt capable in handling the task with the AI's help"). All items but two (marked with * above)

[14]https://ueqplus.ueq-research.org/, last access: 2024-10-18
[15]https://www.ueq-online.org/, last access: 2024-10-23

were measured on a 5-point Likert scale ranging from "strongly agree" to "strongly disagree". The item related to *Interaction Dynamic* also used a 5-point Likert scale but was presented in form of a semantic differential (thus scale ranging from "directive" to "collaborative"). The second item in the *Role Distribution* subscale used a 10-point scale in order to provide us with a more fine-grained distinction between "I contributed more" and "the AI contributed more". Thus, this item is treated individually also in the presentation of results (see Section 5.3).

4.4 Tasks

To capture the essence of the collaborative relationship between humans and AI, the three tasks were carefully designed to replicate the common activities found within each creative field addressed by our study (software development, marketing, UI/UX design). They were conceptualized on the basis of pre-study interviews with practitioners active in these fields, and pre-tested and revised several times before the actual study was conducted. Task design was particularly important for us since the nature of the task has been reported to be a crucial factor also in prior research (see Section 3.4). We thus carefully selected tasks that push participants' creative boundaries, and that encourage them to engage with AI in a way that mirrors real-world collaborative situations. We deemed insights in the interaction process more important than the actual outcome of the tasks. Task complexity was tailored to allow for completion within a 15-minute time frame (which was also approached by several iterations in the pre-testing phase). In all three cases, participants were instructed to consider the given task description as a rough idea and frame, not as a strict guideline. They were told they could add and explore own ideas and should feel free to be creative.

Task 1: Coding with GitHub Copilot. This task required participants to use GitHub Copilot together with an IDE of their choice to implement one of two suggested project ideas. The first idea comprised implementation of an *interactive art generator*, i.e., an application that allows users to create and manipulate shapes (e.g., circles, squares, triangles) interactively, change their size, color, and position through a GUI, and enable exploration of various artistic possibilities. The second project idea required participants to implement an *alarm/timer*, i.e., an application that allows users to set multiple alarms/timers with customizable messages and sounds. The application should provide a user-friendly interface to set, view, and manage alarms/timers. During the programming task, participants should extensively use GitHub Copilot to assist them in writing the code. They were further asked to focus on completing a functional version of the chosen project, include a basic UI to interact with the application, and to provide a documentation for their code.

Task 2: Marketing Text with ChatGPT. This task required participants to use ChatGPT together with a text editing tool of their choice. Participants had to take the perspective of an employee of a hypothetical startup company "TrailTech" whose team is working on the marketing campaign for their latest product, the "Trailrunner Smartwatch", and to work together with ChatGPT to produce an *attractive and persuasive advertisement text* for the website including a one-sentence slogan that works online and on billboard. The participants were further provided with a product information sheet listing the most important features (e.g., water-resistent and shockproof, several sensors for health and fitness tracking, long battery life, real-time weather monitoring, or eco-friendly design using sustainable materials).

Task 3: User Interface Design with Uizard. This task required participants to use Uizard Autodesigner 2.0 to *create a UI design for a user-friendly transportation booking platform*. They had to design 1) an interface where users can easily select a preferred transportation mode and input travel details, 2) a results page that helps users compare different travel options based on criteria such as price, duration, and departure times, and 3) the check-out process.

4.5 Participants

For our study we recruited 15 participants (9 male, 6 female, aged 19 to 53, $M = 32.9$, $SD = 10.8$), distributed equally across three broader creative fields (programming, marketing/writing and UI design, see Section 4.4), resulting in five participants per tool. Our selection criteria emphasized (diversity in) experience in the respective (broader) field, and with AI tools. Tables 4 to 6 list the recruited participants and their respective backgrounds as reported by themselves (except for their ID, which was assigned). The participants reported their previous experience with AI (generally and with the specific tool they used in the study) on a 5-point Likert scale (e.g., responding to: " I have prior experience with GitHub Copilot" using a set of options from 1–"Strongly Disagree" to 5–"Strongly Agree"). We recruited the participants via our university and professional networks, and via their companies.

5 Analysis and Findings

This section presents and analyses the results of our study, structured according to the major elements of our questionnaire as described in Section 4.3: *UX*, *Technology Acceptance*, and *Perception of Collaboration*. Further, we summarize additional findings mainly collected through our observation.

5.1 User Experience

For the analysis of the selected UEQ+ modules (see Section 4.3), we used the data analysis tool that comes with the standardized questionnaire and can be downloaded from its website[16]. It allows for calculation of mean scores and their visualization for all included dimensions, see Figure 2. The items are scaled from -3 (horribly bad) to +3 (extremely good). According to the UEQ's description, scores usually do not exceed +2 or fall below -2 (thus, the *efficiency* dimension reported for ChatGPT in Figure 2b can be regarded considerably high). The error bars show the 95% confidence intervals of the subscale mean. In addition to the box plots, we visualize the findings in radial charts with a colored line for each participant and a black line showing the group median, providing a more granular view on the data (see Figure 3).

ID	Gender (Age)	Profession	Professional Experience	AI Experience (General|Specific)	Usage Freq. (Copilot)	Experience Background and Usage Habits
T1_P1	male (49)	Business Analyst	4 years	5 | 1	never	prior experience using AI systems like ChatGPT, has never actually used it for specific tasks
T1_P2	male (42)	Software Engineer	25 years	5 | 2	rarely	highly experienced with AI systems including ChatGPT and PyTorch, but only uses AI rarely for coding tasks in large projects and as a replacement for Google and StackOverflow
T1_P3	female (38)	Software Developer	1.5 years	3 | 2	occasionally	moderately experienced with AI systems such as ChatGPT, Microsoft Copilot, and GitHub Copilot, occasionally using AI for desktop applications
T1_P4	male (25)	Requirements Engineer	2 years	4 | 1	never	moderate experience level with AI systems like ChatGPT, though he has never used it for specific tasks
T1_P5	male (53)	Software Developer	28 years	3 | 3	occasionally	uses AI systems like Copilot and ChatGPT occasionally for side projects in TypeScript

Table 4. List of participants completing *Task 1 (coding with GitHub Copilot)* including their demographic and professional backgrounds.

[16]https://ueqplus.ueq-research.org/, last access: 2024-10-23

ID	Gender (Age)	Profession	Professional Experience	AI Experience (General|Specific)	Usage Freq. (ChatGPT)	Experience Background and Usage Habits
T2_P1	male (50)	graphic designer	27 years	1 | 2	rarely	has limited experience with AI systems
T2_P2	male (38)	technical editor	6 years	5 | 5	occasionally	frequently uses AI systems like ChatGPT for a variety of tasks including fun projects, work, student projects, creating code, and thesis assistance
T2_P3	female (19)	journalism& strategic communication student	1 year	4 | 4	frequently	frequently uses multiple AI systems, e.g., Perplexity, ChatGPT, Trint, and Elicit for tasks such as homework, presentations, research, leisure activities planning
T2_P4	male (31)	copywriter	1 year	4 | 4	frequently	frequently uses AI systems like ChatGPT and Microsoft Copilot for writing claims, ads, various texts, using it as a search engine alternative to Google, and for brainstorming
T2_P5	female (24)	product& fashion designer	1 year	4 | 4	frequently	frequently uses ChatGPT for writing product texts, concept texts, and presentations

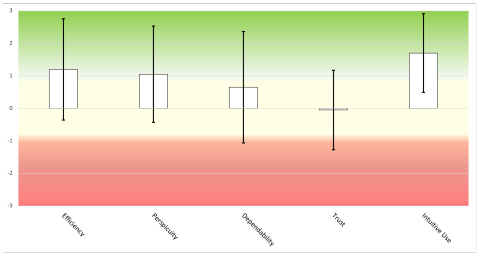
Table 5. List of participants completing *Task 2 (marketing text with ChatGPT)*, their demographic and professional backgrounds.

ID	Gender (Age)	Profession	Professional Experience	AI Experience (general|specific)	Usage Freq. (Uizard)	Experience Background and Usage Habits
T3_P1	female (27)	UI/UX designer	2 years	5 | 1	occasionally	extensively familiar with AI systems like ChatGPT, Midjourney, and Adobe AI; occasionally uses these AI tools for tasks such as changing images, colors, editing site features and sections
T3_P2	male (25)	UI/UX engineer	3 years	2 | 1	never	somewhat familiar with AI systems such as Microsoft Copilot and Adobe Generative AI; has not used these tools for specific tasks yet
T3_P3	male (26)	UI/UX designer	2 years	2 | 1	never	familiar with AI systems like ChatGPT and Suno; has not applied these tools to specific tasks yet
T3_P4	female (24)	UX engineer	1.5 years	3 | 1	occasionally	fairly experienced with AI systems like ChatGPT and Canva AI; occasionally uses these tools but has not specified tasks
T3_P5	female (23)	UI/UX designer	1.5 years	4 | 1	never	highly familiar with AI systems, primarily using ChatGPT and tools from Notion, Miro, Canva; has used these for user studies but not regularly

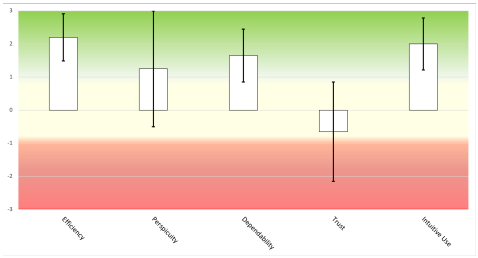
Table 6. List of participants completing *Task 3 (UI design with Uizard)* including their demographic and professional backgrounds.

The use of the median as a measure of central tendency in the analysis is important, as it offers a robust reflection for this small sample size, in contrast to the mean, which can be skewed by outliers. In the following, we discuss the findings related to *UX* for the three tools *GitHub Copilot*, *ChatGPT*, and *Uizard*, and gathered through our three tasks.

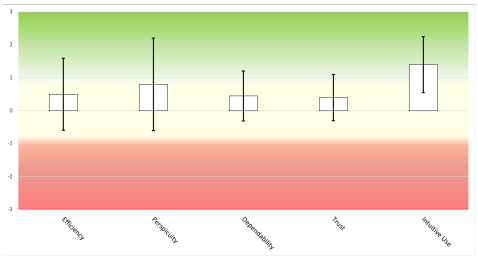
GitHub Copilot. Figure 2a shows that for GitHub Copilot, the subscales *Efficiency* and *Perspicuity* were rated positively but just above the neutral range on the scale. *Dependability* was rated just slightly lower, while *Trust* received the lowest score (a neutral rating), with the confidence interval spanning from negative to positive. *Intuitive Use* was rated relatively high, as evidenced by the error bar, which is concentrated within the positive range. The 95% confidence interval for all values is notably wide. Figure 3a then offers an explanation for this as it shows a high variance between the



(a) Mean scores for *Task 1 (coding with GitHub Copilot)*

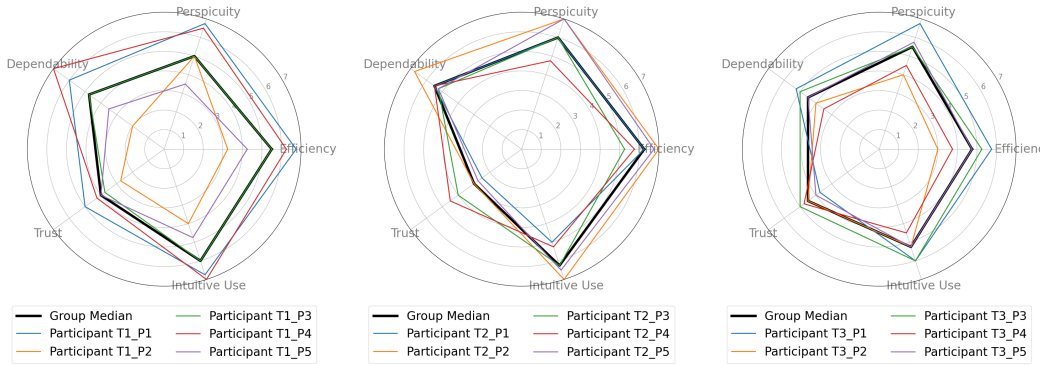


(b) Mean scores for *Task 2 (marketing text with ChatGPT)*



(c) Mean scores for *Task 3 (UI design with Uizard)*

Fig. 2. UEQ+ results gathered for the three tasks (using GitHub Copilot, ChatGPT, and Uizard), visualized as box plots, using the UEQ's standard analysis tool.



(a) GitHub Copilot (b) ChatGPT (c) Uizard

Fig. 3. UEQ+ results gathered for the three tasks (using GitHub Copilot, ChatGPT, and Uizard), visualized as radial charts.

individual participants' scores: participants T1_P2 and T1_P5 in particular being well below the median, and participants T1_P1 and T1_P4 showing higher values. Participant T1_P3 therefore provided the median value for all dimensions.

ChatGPT. Figure 2b shows that for ChatGPT, *Efficiency* and *Intuitive Use* were rated highly as evidenced by narrow error bars concentrated within the positive range. *Perspicuity* and *Dependability* were rated rather positive, with the error bar for *Perspicuity* spanning wide (very high to below neutral) and the error bar for *Dependability* being relatively narrow and in the positive range. *Trust*

being a real outlier, received ratings below zero and an error bar ranging very far to the negative side of the scale. It appears surprising that the participants were seemingly highly content with the tool's efficiency and intuitivity while apparently not fully trusting it. Figure 3b reports a consistent level of satisfaction across all participants, with the median line closely matching individual ratings. While participants T2_P3 and T2_P4 both reported lower scores in most dimensions, they have slightly higher than median scores for *Trust*, which suggests that the group of participants generally had varying impressions on the role of trust in the context of UX.

Uizard. Figure 2c shows mean scores close to neutral for all subscales, indicating a rather low overall result. The only value sticking further into the positive area is *Intuitive Use* with its error bar being almost completely in the positive range. For all other values the error bars range from slightly positive to below neutral. Figure 3c shows an even pattern with all participants tending in the same direction without major outliers, only variance in strength of expression.

5.2 Technology Acceptance

The data gathered via the selected subset of the TAM3 is presented in Figure 4 using boxplots. While the main aim of this visualization is not to directly compare the tools (this is prone to over-interpretation given the small sample sizes), it should rather allow for a general overview of each individual tool's technology acceptance score and facilitate an overview at a glance. In addition to the boxplots, and similar to our reporting for UX, the findings are presented in the form of radial charts to show the scores of each participant individually and provide more detailed insights in Figure 5. In the following, we discuss the findings related to *Technology Acceptance* for the three tools *GitHub Copilot*, *ChatGPT*, and *Uizard*, and gathered through our three tasks.

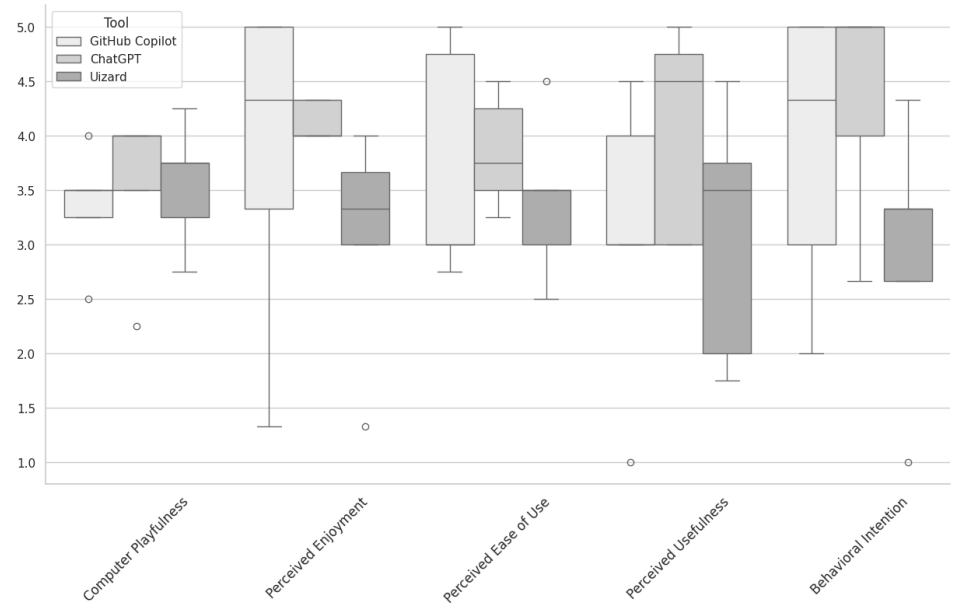
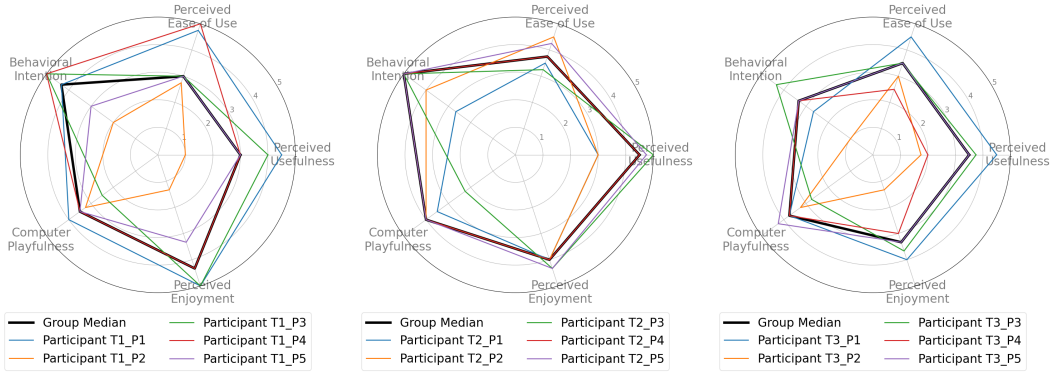


Fig. 4. Overview of TAM3 results gathered for the three tasks (using GitHub Copilot, ChatGPT, and Uizard), visualized as box plots.

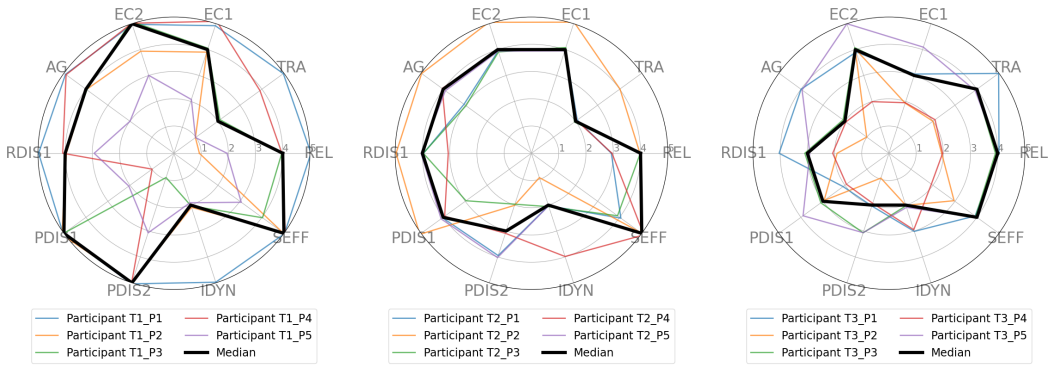


(a) GitHub Copilot (b) ChatGPT (c) Uizard

Fig. 5. Overview of TAM3 results gathered for the three tasks (using GitHub Copilot, ChatGPT, Uizard), visualized as radial charts.

GitHub Copilot. Figure 4 shows a narrow Inter-Quartile Range (IQR) for *Computer Playfulness*. In contrast, *Perceived Enjoyment* and all the tool-related constructs have a very wide IQR. The ratings for *Perceived Ease of Use* indicate that users generally found GitHub Copilot to be easy to use, but there is considerable variability indicating that a few users had challenges. The results for *Perceived Usefulness* indicate that users generally perceived GitHub Copilot as useful. The results for *Behavioral Intention* indicate a strong intention to continue using GitHub Copilot among most users. According to the TAM3 model [54], *Behavioral Intention* is directly influenced by *Perceived Usefulness* and indirectly by *Perceived Ease of Use*. Given the generally moderate perceptions of both *Perceived Usefulness* and *Perceived Ease of Use*, it is interesting that users are inclined to continue using the tool. The variability in the IQR suggests that the majority are positive about future use. Adding a more fine-grained view, Figure 5a shows an uneven pattern with all participants for the tool-related constructs, especially participants T1_P2 and T1_P5 lying well below the median, showing lower values for *Behavioral Intention* and *Perceived Enjoyment*. Participants T1_P1, T1_P3 and T1_P4 show higher values for *Behavioral Intention* suggesting they want to further use the tool.

ChatGPT. Figure 4 shows a narrow IQR for *Computer Playfulness* and *Perceived Enjoyment*, indicating consistent responses across participants, with *Perceived Enjoyment* being rated rather high compared to the other tools. The ratings for *Perceived Ease of Use* show variability, but also in a relatively high range. This variability suggests that while the system is generally perceived as easy to use, some users may have encountered difficulties. Accordingly, Figure 5b shows that participants T2_P1 and T2_P3 rated *Perceived Ease of Use* lower and also have lower ratings for *Computer Playfulness*. *Perceived Usefulness* in Figure 4 shows that users generally perceive ChatGPT as useful, but with high variability. The scores for *Behavioral Intention* indicate a strong intention to continue using ChatGPT among most users. According to the TAM3 model [54], *Behavioral Intention* is influenced by *Perceived Usefulness* and *Perceived Ease of Use*, and despite the variability in these constructs, users seem inclined to continue using ChatGPT. Figure 5b shows that most participants gave similar ratings with high values for *Behavioral Intention* and *Perceived Usefulness*. Participants T2_P1 and T2_P2 show a slightly different pattern indicating they do not perceive the tool as useful as the other participants did (an interpretation that is supported also by the lower values for *Behavioral Intention*).



(a) GitHub Copilot (b) ChatGPT (c) Uizard

Fig. 6. Overview of results related to *Perceived Collaboration* gathered for the three tasks (using GitHub Copilot, ChatGPT, and Uizard), visualized as radial charts. Abbreviations are used according to the description in Section 4.3.

Uizard. We can report results for *Computer Playfulness* similar to the other tools with a bit more variability, according to Figure 4. With *Perceived Enjoyment* being rather low, this could be a factor contributing to the low *Perceived Ease of Use* ratings, according to the TAM3 model. The IQR for *Perceived Ease of Use* is narrow indicating agreement among participants. *Perceived Usefulness* also shows similar findings but with a wide IQR, suggesting that while some users found Uizard useful, others did. The results for *Behavioral Intention* reveal a general low intention to continue using Uizard, with a narrow IQR. This suggests that users' future intentions are closely linked to their immediate experiences and perceptions of the tool. Figure 5c also shows mixed reactions for Uizard. While, as already mentioned, all the scores are generally low, it can be seen that participants T3_P1 and T3_P3 gave high values for *Perceived Usefulness* and *Perceived Ease of Use*. While most users provided neutral scores for *Perceived Enjoyment*, they also gave neutral scores for *Behavioral Intention*. Participant T3_P2, with very low scores for *Perceived Usefulness* and *Perceived Enjoyment*, also gave a very low score for *Behavioral Intention*, indicating no intention to further use the tool.

5.3 Perception of Collaboration

The specific questions we added to the general ones and standardized instruments, capturing participants' perception of the interaction process in terms of a "collaboration", are analyzed and visualized in radial charts with the group median values (see Figure 6). In this case, we refrained from an additional visualization in form of box plots (as used for UX and Technology Acceptance). The reason for this is that here, we do not have established constructs but analyze on the basis of individual items. All items and their abbreviations are described in detail above in Section 4.3. As introduced there, most items use a 5-point Likert scale ranging from 1–"strongly disagree" to 5–"Strongly agree". Exceptions are *Interaction Dynamic (IDYN)* (also using a 5-point scale but ranging from 1–"Directive" to 5–"Collaborative"), and *Role Distribution*, second item (*RDIS2*) ranging from 1–"The AI contributed more" to 10–"I contributed more". Due to the divergent scale, and because we consider *RDIS2* a key item, the related results are presented separately in Figure 7. In the following, we discuss the findings related to *Perception of Collaboration* for the three tools *GitHub Copilot*, *ChatGPT*, and *Uizard*, and gathered through our three tasks.

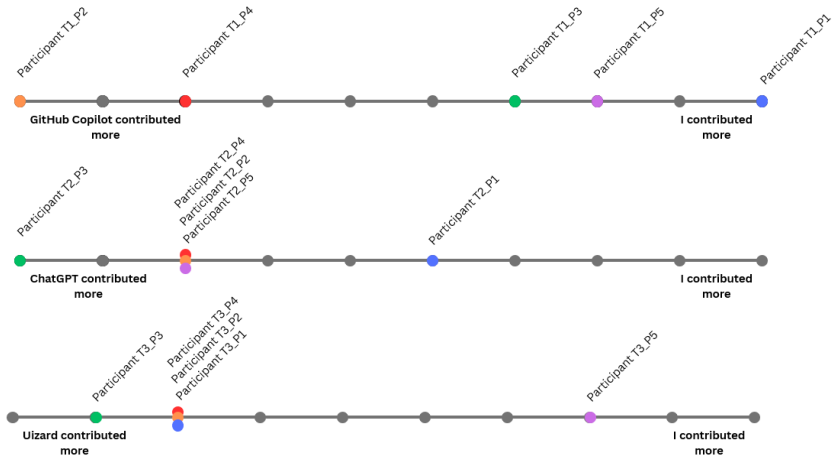


Fig. 7. Participants' responses to *RDIS2: Who contributed more?* for GitHub Copilot (top), ChatGPT (middle), and Uizard (bottom).

GitHub Copilot. In general, Figure 6a shows high variance in scores, with participants T1_P2 and T1_P5 rating below median for most items, while T1_P1 and T1_P4 have higher values. This is especially apparent for *Transparency (TRA)* and *Trust and Reliance (REL)*, indicating that participants T1_P1 and T1_P4 understood how the AI made its decisions and felt confident about the AI's ability. The median score of *Interaction Dynamic (IDYN)*, which is 2, indicates that the users felt the interaction to be more of directive than collaborative nature. In supplement to this, Figure 7 (top) reveals a broad range: while participants T1_P2 and T1_P4 reported that GitHub Copilot contributed more to the task, the other participants stated that their own contribution was more.

ChatGPT. Figure 6b shows low scores for *Transparency (TRA)*. The other items such as *Self-Efficacy (SEFF)* received overall high scores, indicating the users felt capable in handling the task with the AI's help. The results for *Interaction Dynamic (IDYN)* but also *Role Distribution (RDIS2)*, see Figure 7 (middle), however further suggest that the participants perceived the interaction as rather directive than collaborative, and that the AI contributed more (with exception of participant T2_P1 who reported to have contributed slightly more than ChatGPT).

Uizard. Generally, the results for Uizard (see Figure 6c) seem to be lower than for the other tools, and additionally show a high variance between users. Specifically, the results for *Interaction Dynamic (IDYN)* and *Role Distribution (RDIS2)* indicate that the interaction felt highly directive and like the AI contributed more, see Figure 7 (bottom), with the notable exception of participant T3_P5. The low scores for *Aligned Goals (AG)* indicate that users did not feel their own goals and the goals of the AI to be aligned. In contrast, the comparatively higher median values for *Transparency (TRA)*, *Trust and Reliance (REL)*, and *Effective Communication (EC2)* suggest that users understood how the AI made its decisions, felt confident about the AI's ability and the its communication to be relevant.

5.4 Additional Findings

We complement the results reported above in Sections 5.1 to 5.3, which are based on participants' responses to our exhaustive questionnaire, by additional findings on the basis of observations (and partly, individually tailored brief interviews). Further, a thorough discussion linking these

results to the results of the questionnaire in order to allow for more in-depth and comprehensive interpretation of the available data, can be found in Section 6.

Participant	Interaction Modality	Communication Style	Description	Exemplary Phrases
T1_P1	chat only	6	Polite but direct	I want to... please make...
T1_P2	inline and chat	3	Direct, task-focused	create, use, add...
T1_P3	inline only	2	Minimal, very formal	an object..
T1_P4	inline only	5	Balanced, polite	I need a, I want...
T1_P5	chat only	3	Direct command, not polite	add...
T2_P1	chat only	2	Repetitive, formal	generate...
T2_P2	chat only	8	Conversational, friendly	Hi there... Alright...
T2_P3	chat only	3	Task-focused	write me, write it...
T2_P4	chat only	8	Polite, slightly formal	I need..., please...
T2_P5	chat only	9	Friendly, conversational	hello, I need, thanks...
T3_P1	text prompts only	3	Task-focused, formal	design...
T3_P2	text prompts only	3	Task-focused, formal	design... include...
T3_P3	multiple modes	3	Task-focused, formal	design... create...
T3_P4	text prompts only	3	Task-focused, formal	design... create...
T3_P5	multiple modes	2	Very formal, minimal	app...

Table 7. Participants' interaction modalities and communication styles with phrases and justifications, grouped by tool/task.

5.4.1 Interaction Modality. To discuss interaction modality, it is necessary to first distinguish between the capabilities of the individual tools. Although ChatGPT generally offers multiple input and output capabilities, it was only used for text-based interaction in this task. GitHub Copilot can mainly generate code and also provide text responses. It takes input via chat with text or code snippets, but can also be used inline, directly in the code. Our participants have used the features in different ways: some used only the chat functionality, others used both, or only the inline functionality. Uizard offers many input options, with the main modes of interaction being selection on the GUI, and text input. Most participants used only these options, however others also used additional ones, such as the "focus predictor" or the "theme generator". Table 7 lists the participants' interaction modalities used.

5.4.2 Communication Patterns. We could observe varying communication patterns across our 15 participants using the three AI tools: while some users were very friendly and communicated with the AI more like it was a *peer*, others were exceptionally strict and formulated precise instructions in order to just *steer*. Particularly notably, some users used phrases such as: "Fix it please" (T1_P2), "Alright, we have a..", "All right, all good" (T2_P2), "Hello, I need", "Thanks" and "More creative pls" (T2_P5). In Table 7, we, besides interaction modality, report a "classification" (using a 10-point scale ranging from 1–"strict/formal" to 10–"friendly/informal") and general description of the communication style, and list exemplary phrases. Please note that this interpretation has been made by the authors based on observation data, not by the participants themselves. While not representative due to the small sample size and deliberate heterogeneity within the three groups (and among participants), we can see a tendency towards more informal communication with tools that aim to act in a most natural way themselves (e.g., through natural-style language), and that offer a broader range of functionalities that make it necessary to communicate more intensively about, and more formal communication with tools that have a more narrow set of functionalities and that target a highly specific field of use (in our case, this best applies for Uizard). This is not only suggested by the communication style scores listed in Table 7, it is also clearly indicated by

the participants' exemplary phrases. Regarding the scores, the mean value for GitHub Copilot is 3.8 (ranging from 2 to 6 with a median of 3), for ChatGPT it is 6.0 (ranging from 2 to 8 with a median of 8) and for Uizard it is 3.0 (ranging from 2 to 3, with a median of 3). Further interesting observations related to communication style are aspects of personification and embodiment. As described in Section 3, Rezwana et al. have shown in their study how embodiment can have positive effects on collaboration [44]. In our data, we could identify several user statements across all three tools which personified the AI, referring to it as if it were a human. For instance, participant T1_P5 remarked "Let's see if he knows,..", participant T2_P3 noted "He did a good job!" and participant T3_P1 observed "Sometimes he just does whatever." (note the personal pronoun "he" instead of "it" the participant used to refer to the AI).

6 Discussion

In the previous sections analyzing our findings on the basis of the questionnaire but also our observations and brief interviews on occasion, we already identified a lot of interesting "patterns" related to overall perception of the interaction with AI tools, potentially suggesting more general effects. Our pilot study, whose design and methodology was informed by the theoretical framework we derived from the related literature in the form of relevant *core concepts* (see Section 3), was not intended to lead to robust evidence for such effects, but to raise concrete research questions and potential hypotheses. Further, it was not intended to offer a direct comparison of tools (an endeavor that would require a different study design), but to discuss different human-AI settings individually. In the following, we reflect on our observations and major conclusions further – offering a basis for future research but also linking them to existing research (including but not limited to the papers in our corpus of literature described in Section 3). One main aspect that inspired and directed our research evolved around the perception of human-AI interaction by humans – in terms of whether such a setting is perceived as "collaborative" (the AI is seen as *peer*) or "directive" (the AI is seen as *tool* the human can *steer*). We expected a number of factors to potentially contribute to this perception – including *UX*, general aspects around *Technology Acceptance*, but also collaboration-specific aspects such as *Role or Power Distribution*.

6.1 Domain-Specific Dynamics and Level of Experience

Informed by related literature, which suggests that experts and non-experts might be affected by involvement of AI in the creative process in different ways (see the discussion in Section 3.1), we tried to recruit participants with a varying level of domain expertise. For the first two groups of participants (*Task 1* and *Task 2*), we could achieve a higher variation than for the five participants in *Task 3*. The results for the programming task (*Task 1*) actually suggest that there might be such a difference as the two highly experienced participants (T1_P2 and T1_P5 with 25 and 28 years of experience, respectively) show a similar tendency towards lower scores for pragmatic quality (cf. the UEQ+ results) and partly also technology acceptance (cf. the TAM3 results) than the less-experienced participants. For the marketing text task (*Task 2*), where there is one participant with considerably higher level of domain experience (T2_P1) compared to the others, this potential effect is less obvious but still present (primarily for the technology acceptance scales). For the UI design task (*Task 3*) where the group was generally less diverse regarding experience, we could still note a tendency in the same direction: the participant with the highest level of experience (T3_P2) provided the lowest scores for a majority of the scales (e.g., *Perspicuity* and *Efficiency* and all but one TAM3 scale). Thus, we suggest a hypothesis that confirms and exceeds Hwang's discourse on push-back from human domain experts: ***H1: Co-creative AI is more helpful for and better accepted by less experienced users, independent of the concrete target domain.***

The role of a human user's expertise in the context of human-AI co-creativity is also emphasized in Moruzzi and Margarido's conceptual framework. They suggest that users should be able to "communicate to the AI their familiarity and confidence with the process they are going to perform in coordination" [37]. A related explanation for a correlation between level of human experience and acceptance of co-creative AI could be that most current AI systems have been notoriously designed for non-experts as also hinted in [21] (i.e., not sufficiently adapting to users' individual level of expertise as suggested in [36, 37]).

6.2 Role Distribution and Interaction Dynamic

In Section 5.3, we already discussed *Interaction Dynamic (IDYN)* and *Role Distribution* (specifically *RDIS2* which is reported in Figure 7) on the basis of the questionnaire. The findings suggest a connection between these two items in form of the following pattern: a low value for *RDIS2* often co-occurs with a low value for *IDYN*. This means that users who felt that the AI had contributed more, also felt the interaction to have been more directive than communicative. An exemplary related quote of participant T1_P4 underlines this: "In my work as a requirements engineer I just tell people what to do – feels the same here" and is echoed in his choice of words with phrases such as: "I want...". Interestingly, *RDIS1 ("the distribution of roles between me and the AI was effective for the task")* at the same time is considerably higher for this participant (similarly, also for others), which indicates that they still felt the role distribution to be adequate for the task. This again implies that balanced role distribution but also a less directive communication style is no strict prerequisite for successful and satisfactory solving of tasks. For instance, in the case of T1_P4, this further becomes evident when comparing to the results of the UEQ+ (where the participant rated *Efficiency*, *Dependability* and *Perspicuity* – all subscales related to the pragmatic quality of a tool [27] – to have been high). Based on our observation, we formulate the following hypotheses (H2 is two-sided thus we consider and phrase it in two variants): ***H2_1: The lower the human's perceived level of own contribution, the more the communication style is perceived as "directive" rather than "communicative"***, ***H2_2: The more the perceived communication style with an AI leans towards "communicative" rather than "directive", the higher users will rate their own contribution***, and ***H3: There is no correlation between communication style and task efficiency in human-AI collaborative interaction.***

Linking these hypotheses to existing research, we point to Rezwana and Maher's framework for modeling interaction in human-ai co-creative systems [41, 43]. They emphasize the significance of communication and communication style for perception of an AI agent as "intelligent entity and a co-equal creative partner rather than a mere tool" and the overall co-creation as more balanced [43]. This supports the assumption that communication between humans and AI might follow similar basic concepts of communication, such as grounding. "Grounding", in the understanding of Clark and Brennan, can be described as the incremental, collaborative process by which communicators establish mutual understanding in a conversation [9]. Further, the relevance of communication style has recently been discussed by Joshi and Bengler, who propose that adopting a communicative interaction style, which fosters collaboration and two-way dialogue, is more likely to create a participatory experience. In contrast, a directive style, which focuses on control and providing instruction, might limit user engagement and reduce participation in the interaction [24]. This seems to fit well into the scope of our hypotheses H2_1 and H2_2 since it could be assumed that high user engagement also relates to a user's sense of contribution. Regarding H3, there appears to be evidence suggesting that multiple factors, such as user expertise and familiarity with AI [40], as well as task complexity [45], may influence task efficiency in human-AI interaction. However, to date, we have not identified previous research that explicitly points to a direct correlation between communication style and task efficiency in this context.

6.3 Power Distribution

Indirectly related to *Role Distribution* and *Interaction Dynamic* as discussed above, is the concept of *Power Distribution*. Here, we can note considerably high (median) values for both related items, *PDIS1 ("I felt in control of the decisions made during the task")* and *PDIS2 ("the decision-making power seemed balanced between me and the AI")* for GitHub Copilot (see Figure 6a), suggesting that users, although they perceived the communication as rather directive, still perceived the decision-making power as balanced. Interestingly, this is particularly the case for those participants who felt that the AI had contributed more to the result (T1_P2 and T1_P4), according to Figure 7 (top). Two of three participants who felt that they had contributed more (T1_P3, T1_P5, and T1_P1) did *not* perceive power distribution as balanced, however T1_P1 and T1_P3 still had the feeling they had control over the decisions made during the task. Related to the other tools (most prominently, for Uizard), the items linked to *Power Distribution* were generally rated lower, suggesting that participants felt less in control of the decision making process (cf. *PDIS1*) on one hand, and they perceived the power distribution as less balanced (cf. *PDIS2*) on the other. Again, a possible reason for this could be rooted in the nature of the respective task. In coding (*Task 1*), it is more often relatively straight-forward what the result should look like, whereas this is not entirely true for generating marketing texts (*Task 2*) or screen designs (*Task 3*). Especially in case of the latter, there is a lot of creative freedom and the possibilities of design choices are seemingly endless. This could on one hand suggest that the more creative a task is in the sense of offering more freedom, the more users feel like they are losing control and decision making power. Relatedly, the participants working with Uizard made interesting statements such as: "Sometimes it does whatever" (participant T3_P1) or "It didn't work the way I imagined it would" (participant T3_P4). On the other hand, reasons could also be related to the communication modalities and styles the tool itself offers, and to its general scope (the broader the range of functionality, the more communication is necessary to find a solution for a concrete task, which again might contribute to users feeling in control of the process). In conclusion, we suggest the following two (connected) hypotheses based on our findings: ***H4: Human's sense of control in human-AI co-creative settings is dependent on the nature and complexity of the task (the less straight-forward the task, the less the sense of control)***, and ***H5: Human's sense of balanced power distribution in human-AI co-creative settings is dependent on the nature and complexity of the task (the less straight-forward the task, the less balanced power distribution is perceived).***

Related to task complexity, Salimzadeh et al. describe multiple "task characteristics" contributing to it (including information load, information diversity or rate of information change) [45]. They further point to previous literature that thoroughly discusses task complexity (although not in the context of human-AI interaction). Specifically, Liu and Li suggest a task complexity model considering task structure, resource requirements, and interaction between task and performer characteristics. They list multiple related "complexity contributory factors", including unstructured guidance diversity, inaccuracy, non-routine events, or heterogeneity [32]. This understanding greatly fits with our observation that tasks allowing for and requiring more creative freedom (specifically, Task 3 in our study) were perceived as more complex. Further, it also fits well with our H4, since Liu and Li also argue that, e.g., unpredictability (which goes hand in hand with a weaker sense of control), contributes to complexity. Salimzadeh et al. argue in their (semi-systematic) literature review that most of the empirical studies on human-ai decision making, have used low-to-medium complexity tasks [45]. This suggests that more empirical research, more strongly varying complexity of tasks is needed (which fits well with our hypotheses H4 and H5). Further, Iftikhar et al., who explore human-agent team dynamics present a conceptual teams framework, suggest a connectivity between "team structure and structural contingencies", and "task structure"

(explicitly including scope and complexity), which they both consider "structural factors" [23]. In addition, they identify "compositional factors" (e.g., team member characteristics) and "mediating mechanisms" (e.g., information sharing or team processes). These components, according to Iftikhar et al. all contribute to team performance. They point out that there should be a "nuanced balance between agent autonomy and task interdependence" [23] and identify a research gap (among others) related to shared leadership, which strongly relates to our suggested H5 related to power distribution.

6.4 Trust and Transparency

Regarding *Trust and Reliance (REL)* and *Transparency (TRA)* (see Figure 6) but also the *Trust* subscale in the UEQ+ (see Figures 2 and 3), it is noteworthy that especially for GitHub Copilot and ChatGPT, all related scores are relatively low (in particular, those for *Trust* in the UEQ+, but also those for *Transparency* in the collaboration-related part of the questionnaire). In summary, even though Uizard received a bit higher ratings in this context, there seems to be a consistent lower level of trust across all tools and users. For example, participant T1_P1 often wondered whether certain mistakes had been made by the AI or by his own input. Also the participants did oftentimes not seem to be confident that the AI will be able to handle the task (which is particularly surprising because some users distrusted the tool even though they were satisfied with its efficiency), which is further exemplified by statements such as "Let's see if he knows" (T1_P5). Also, the AI's output led to mixed reactions ranging from "He did a good job!" (T2_P3) to "Did what I asked for." (T3_P5) but also "AI won't replace a designer anytime soon." (T3_P4). From our findings we derive that the scope of the AI tool and its communication style have an effect on users' *Trust* and perceived *Transparency*. Based on this, we formulate the following hypothesis for future research: ***H6: There is a negative correlation between a co-creative AI tool's scope and complexity, and users' trust and perceived transparency (the wider the scope and the more complex its capabilities, the less transparent and trustworthy).***

Our suggested hypothesis H6 aligns with existing research on transparency and trust in (explainable) AI systems, and on trust in human-ai interactive systems in specific. Prior evidence indicates that while increased transparency can reduce uncertainty and enhance trust [1, 4, 31, 34, 55, 58], as system complexity increases, achieving transparency becomes more challenging [20, 34, 56]. Furthermore, the "black box" nature of complex AI systems heightens users' concerns about risks and uncertainties, suggesting that increased complexity can negatively impact perceived transparency and trust [6, 20].

7 Limitations and Conclusion

In this paper, we explored the nature of collaboration between humans and AI in three different creative tasks, informed by a theoretical framework derived from a focused systematic literature review, in the form of a pilot study. The main aim behind our work was to identify concrete research questions and hypotheses for future research. From the analysis of our findings, we could derive several interesting patterns that could become subject to individual, targeted quantitative empirical studies in the future. While our study was not intended to be exhaustive on its own, nor to lead to statistically reliable results, we still acknowledge the following limitations.

Small sample size. Our study included a relatively small number of 15 participants in total (5 for each task). We are aware that this naturally affects the robustness of results. We thus did not aim to draw statistically reliable conclusions but to generate well-grounded hypotheses in order to inform future studies with larger samples size, even better controlled conditions and a focus on quantitative measures.

Participant diversity. While the participant groups were diverse as desired mainly for *Tasks 1* and *2*, this is not entirely true for *Task 3* (UI design with Uizard) where we would have wished for higher diversity regarding both *professional experience* and *usage frequency* with the concrete tool, but also *age*.

Task complexity. We acknowledge that the tasks we used throughout our study were all of small scale (in order to fit in the scope of a study), and that this might thus differ from everyday real-world tasks which naturally vary in terms of scope and complexity. However, we tried to mitigate this effect through our thorough pre-tests which led to considerable revision of task definitions, particularly for the programming task. There, a first pre-test revealed that the intended task had not been sufficiently complex (the programmer was able to solve it with one simple interaction with the AI tool) and we varied the task to allow for a higher level of interactivity between human and AI.

Self-reported data. We further acknowledge that the part of our data which relies on self-reporting (i.e., via our questionnaires and the brief interviews), might introduce a potential bias, e.g., due to the participants' experience with the AI tools. We however argue that we carefully selected our instruments and included standardized ones where possible (i.e., UEQ+ and TAM3 items). These have been carefully tested for reliablity and validity, and are thus designed to minimize potential biases (including those introduced through self-reporting). Also, we complemented the self-reported data by data gathered through observation (including interaction with the respective AI tool), and mostly found that one data source confirmed the other.

Focus on three AI tools. In our study, we concentrated on three specific AI tools we consider representative of their respective domain (programming, marketing texting, and UI design). We acknowledge that this is just a small sample of the available AI tools. However, we considered a large number of 51 tools initially, and we carefully assessed them based on our selection of criteria (see Section 4.1), and tested those we considered suitable candidates for our purposes, before we finally selected three for our study. For instance, these thorough tests led to a selection of a different tool for the UI design task because we identified unexpected UX issues with the one we had initially intended to use.

General usability. Finally, the generally lower scores for the pragmatic quality of *Uizard* could somewhat mitigate the significance of the findings in the other dimensions.

Despite these limitations, we would also like to point to the strengths of our research. Specifically, these are: 1) the systematic and focused analysis of related literature, providing 2) a thorough theoretical framework for our own study in form of the *core concepts* presented in Section 3, 3) the wide scope of our study spanning across three different creative fields and tool/task combinations, 4) the diversity of our study participants who had different backgrounds and levels of expertise, 5) the continuity from related literature to study design, methodology and analysis.

Acknowledgments

This research has been partly funded within the scope of the Human-Centered Artificial Intelligence (HCAI) project [DFH 23-N], and the Hybrid Collaboration Spaces (HYCOS) project [P 34928], both funded by the Austrian Science Fund (FWF).

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Appendix: Additional Tables

AI Tool	Interaction Modality	Availability	Complexity	Functional Range
Open AI ChatGPT-4	**Text**	**Subscription-Based**	**Easy**	**Broad**
Open AI ChatGPT-3.5	Text	Free	Easy	Broad
CapCut	GUI, Text	Free	Moderate	Broad
Microsoft Copilot	GUI	Enterprise-Level/Extensive Cost	Moderate	Broad
DeepL Translator	Text	Free	Easy	Specialized
DeepL Write	Text	Free	Easy	Specialized
GrammarlyGo	Text	Free	Easy	Broad
Copy.ai	GUI, Text	Enterprise-Level/Extensive Cost	Difficult	Broad
Wordtune	GUI, Text	Free	Easy	Broad
Writesonic	Text	Subscription-Based	Easy	Broad
Rytr	Text	Free	Easy	Broad
Adobe Firefly	Text	Subscription-Based	Moderate	Broad
Fronty	Image	Free	Easy	Specialized
Fireflies	Multi-Modal	Enterprise-Level/Extensive Cost	Easy	Specialized
Midjourney	Text	Subscription-Based	Easy	Specialized
DALL-E 2	Text	Subscription-Based	Easy	Specialized
Tome	GUI, Text	Free	Easy	Specialized
Lumen5	GUI	Subscription-Based	Easy	Specialized
GitHub Copilot	**Text, Code**	**Subscription-Based**	**Moderate**	**Specialized**
Jasper	Text, GUI	Subscription-Based	Easy	Broad
Podcastle	Voice, GUI	Free/Subscription-Based	Easy	Specialized
Soundraw	GUI, Text	Subscription-Based	Easy	Specialized
DeepArt	Text	Subscription-Based	Easy	Specialized
RunwayML	GUI, Text	Free/Subscription-Based	Moderate	Broad
Artbreeder	GUI	Free/Subscription-Based	Easy	Specialized
Zyro AI Content Generator	GUI, Text	Subscription-Based	Easy	Specialized
Synthesia	GUI	Subscription-Based	Easy	Specialized
Descript	GUI, Text	Subscription-Based	Moderate	Broad
Canva's Magic Write	GUI, Text	Free/Subscription-Based	Easy	Specialized
BrainShop.AI	Text	Free	Easy	Specialized
Beautiful.AI	GUI	Free/Subscription-Based	Easy	Specialized
Visme	GUI	Free/Subscription-Based	Easy	Broad
Rephrase.ai	GUI, Text	Subscription-Based	Moderate	Specialized
Voiceflow	GUI, Text	Subscription-Based	Moderate	Specialized
Crayon	GUI	Subscription-Based	Moderate	Broad

Table 8. List of considered AI tools and assessment based on selected criteria: Interaction Modality (one or several of the following: *GUI, Text, Code, Draw, Multi-Modal*), Availability (*Free, Subscription-Based, Enterprise-Level or Extensive Costs, Not Available*), Complexity (*Easy, Moderate, Difficult*), and Functional Range (*Specialized, Broad*). Items ordered arbitrarily. The tools eventualy selected for the study are highlighted **in bold**. *Please note, that this should be considered a snapshot of the tool landscape at the time the study was conducted. It has been added for reasons of transparency, however, since AI tools rapidly develop and evolve, this list is neither exhaustive nor still up-to-date.*

AI Tool	Interaction Modality	Availability	Complexity	Functional Range
Rosebud AI	GUI, Text	Subscription-Based	Moderate	Specialized
Murf AI	GUI, Text	Subscription-Based	Easy	Specialized
Play.ht	GUI, Text	Subscription-Based	Easy	Specialized
DeepCode	Text, Code	Free	Difficult	Specialized
Kuki Chatbot	Text	Free	Easy	Specialized
BigGAN	Code	N.A.	Difficult	Specialized
QuillBot	GUI, Text	Free/Subscription-Based	Easy	Specialized
AIVA	GUI, Text	Subscription-Based	Moderate	Specialized
Skylum Luminar	GUI	Subscription-Based	Moderate	Specialized
DeepFaceLab	Code	Enterprise-Level/Extensive Cost	Diff	Specialized
Quick, Draw!	Draw	Free	Easy	Specialized
Uizard	**GUI, Text**	**Subscription-Based**	**Easy**	**Specialized**
Pictory	GUI	Subscription-Based	Easy	Specialized
Let's Enhance	GUI, Text	Free/Subscription-Based	Easy	Specialized
Replika	Text, GUI	Free/Subscription-Based	Easy	Specialized
VALL-E	Text	N.A.	Moderate	Specialized

Table 9. (Table 8 ctd.) List of AI tools and assessment based on selected criteria.

Received October 2024; revised April 2025; accepted August 2025